PORTFOLIO PROJECT

iGaming Game Metadata Database & Data Quality

A simulated provider-feed validation and data handover workflow

PROJECT SNAPSHOT

SCOPE	TOOLS	DATASET
Metadata validation, discrepancy investigation, standardisation and handover	Microsoft Excel, Pivot Tables, formulas and reference tables	100 synthetic game records from five named providers

IMPORTANT DATA NOTE

All game records, titles, RTP values, provider mappings and market rules in this project are synthetic training examples. Provider names are used illustratively. This is not an actual provider feed, commercial integration or legal market approval dataset.

Independent learning project

1. Executive summary

This project simulates the operational work involved in receiving game metadata from multiple suppliers, validating the information, investigating discrepancies and preparing standardised records for a central game database.

The exercise began with a 100-record CSV feed containing 12 metadata fields. A reference workbook supplied canonical provider names, provider codes, supported game types, market-code examples and a data dictionary. The work was completed in Excel using filters, conditional formatting, Pivot Tables and formulas.

MEASURE	RESULT
Incoming records	100
Metadata fields	12
Documented issue types	10
Records passing initial automated checks	56
Records held for data-quality review	44
Records eligible for simulated handover	52
Records held for external verification	4
Exception queue size	48

HOW TO READ THE RESULTS

The 56 / 44 split is the outcome of the workbook's initial record-level rules. The later 52 / 44 / 4 handover split applies an additional verification flag. These are record classifications, not counts of unique issue instances. A single record can have multiple issues.

Project objectives

- Understand the incoming feed structure and distinguish identifiers from descriptive attributes.
- Detect duplicates, missing values, invalid values, naming inconsistencies and cross-field conflicts.
- Use reference tables to investigate and propose corrections without overwriting source data.
- Revalidate the working dataset and separate records ready for handover from exceptions.

2. Data model and field overview

Each row represents one game record. The game identifier is intended to uniquely identify a game within the simulated central database. Other columns describe the game and its operational attributes.

FIELD	MEANING	VALIDATION FOCUS
game_id	Unique game identifier	Uniqueness; provider-code consistency
game_name	Displayed or internal game title	Presence; identity consistency
provider	Game supplier/provider	Canonical name; reference mapping
product_type	Broad product grouping	Approved category
game_type	Specific game format/category	Approved values; provider compatibility
rtp_percent	Theoretical return-to-player percentage, where applicable	Numeric format; range; applicability
volatility	Outcome-variance classification, where applicable	Allowed values; applicability
features	Game mechanics/features	Consistent labels; multiple values
release_date	Game release date	Valid calendar date; parseable format
market_codes	Markets associated with the game	Validate each code independently
mobile_supported	Mobile support indicator	Consistent Yes/No representation
status	Operational status	Approved status values

Important distinctions

- Identifier vs measurement: game_id is a key, not a quantity to calculate.
- Product type vs game type: product_type is broad; game_type is more specific. In this training feed, all records have product_type = Casino, while game_type varies.
- Feature vs theme: features describe mechanics or bonus elements; a theme describes presentation or setting.
- Blank vs N/A: blank means no value supplied; N/A is an explicit representation that a field may not apply. Applicability must be defined by the data specification.

3. Validation workflow

STAGE	CHECKS PERFORMED	OUTPUT
1. Inspect	Record count, columns, data types, distinct categories	Data structure understood
2. Validate identifiers	Duplicate game_id values	Duplicate issue recorded
3. Validate completeness	Blank RTP, provider, game_type and volatility fields	Missing-value issues recorded
4. Validate formats	RTP scale, date strings, accepted labels	Format and invalid-value issues recorded
5. Validate references	Provider names, provider codes, market codes	Reference mismatches flagged
6. Validate relationships	Provider/game-type compatibility; identifier/name/provider alignment	Cross-field exceptions identified
7. Clean working copy	Separate cleaned/proposed columns; preserve raw input	Traceable cleaned dataset
8. Revalidate	Repeat checks against cleaned/proposed fields	PASS / HOLD classification
9. Prepare handover	Separate candidate dataset and exception queue	Handover-oriented outputs

Principle: preserve the source

The incoming feed was treated as immutable. Proposed corrections were placed in separate working columns, and unresolved values were retained as exceptions rather than guessed. This provides traceability between the source value, proposed value, evidence and verification status.

OPERATIONAL RULE

A validation rule can identify a suspicious value. It cannot, by itself, prove what the correct value should be. Use authoritative provider documentation or an approved internal reference to verify corrections.

4. Issue register

Ten issue types were documented during the exercise. Some issues affect the same record; counts should not be added to infer the number of unique affected games.

ID	ISSUE	FINDING	SCOPE	STATUS
DQ-001	Duplicate game_id	HG_0010 appears on two different games from different providers.	2 records	Pending verification
DQ-002	Missing RTP	40 records have an empty rtp_percent value. Applicability needs confirmation.	40 records	Investigating
DQ-003	Missing provider	PP_0089 has a blank provider; the PP prefix and title suggest Pragmatic Play.	1 record	Pending verification
DQ-004	Missing volatility	PP_0001 is a slot game with a blank volatility value.	1 record	Open
DQ-005	Provider mismatch	EV_0036 has an EV identifier and Evolution title but provider value PragmaticPlay.	1 record	Pending verification
DQ-006	Missing game type	HG_0043 has no game_type value.	1 record	Open
DQ-007	RTP scale inconsistency	HG_0028 has 0.9624 while other values use a percentage scale.	1 record	Pending verification
DQ-008	Invalid date	BG_0078 contains the impossible date 31/31/2024.	1 record	Open
DQ-009	Invalid market code	BG_0054 contains EE;GB;XX; XX is not in the simulation reference.	1 record	Investigating
DQ-010	Inconsistent game-type label	HG_0067 uses Slots instead of the canonical label Slot.	1 record	Resolved in working copy

5. Investigation notes and proposed actions

Identifier and provider discrepancies

DQ-001: Two records from Hacksaw Gaming and Play'n GO share HG_0010. The Provider Reference maps HG to Hacksaw Gaming and PG to Play'n GO. PG_0011 was proposed for the Play'n GO record because its title identifies it as demo game 011. This is a candidate correction only; provider confirmation or an authoritative identifier mapping is still required.

DQ-003: PP_0089 has a blank provider. Its identifier prefix and game title suggest Pragmatic Play. Pragmatic Play was entered as a provisional cleaned value, retaining the original blank value and documenting the evidence.

DQ-005: EV_0036 has an Evolution ID prefix and an Evolution game title, but its provider field says PragmaticPlay. Evolution was proposed as the provider value, subject to source verification.

Missing and ambiguous values

DQ-002: The 40 blank RTP records were distributed across the received provider labels: Evolution 20, BGaming 12, Pragmatic Play 7 and PragmaticPlay 1. The two Pragmatic labels refer to a spelling variant in this exercise. The missing-RTP group comprises 23 Table Game, 16 Live Casino and 1 Virtual Sports record. All 40 also have N/A volatility. This correlation does not prove that RTP is inapplicable; provider specifications must determine that.

DQ-004 and DQ-006: PP_0001 has missing volatility; HG_0043 has missing game type. Neither value was inferred from unrelated fields.

DQ-007: 0.9624 × 100 = 96.24. The result is consistent with the observed Hacksaw sample range of 94.24–97.76, but the provider's original RTP representation remains the authoritative source.

Format and reference issues

DQ-008: Valid release dates arrived as ISO-style text (YYYY-MM-DD), which is normal in a CSV import. They were converted to Excel date values in a separate working column. Only the impossible value 31/31/2024 was treated as invalid.

DQ-009: BG_0054 contains EE;GB;XX. The formula split the semicolon-separated list and validated each code against the reference values EE, GB, MT and CW. XX was retained as an exception.

DQ-010: HG_0067 used Slots. It was standardised to Slot in the working copy using the reference category.

6. Provider-level completeness analysis

A Pivot Table was used to examine the 40 missing RTP values by provider label as received. The denominator matters: comparing raw counts alone can make providers with more supplied games appear worse.

PROVIDER LABEL AS RECEIVED	TOTAL RECORDS	MISSING RTP	MISSING RATE
BGaming	24	12	50.0%
Evolution	20	20	100.0%
Play'n GO	20	0	0.0%
Hacksaw Gaming	19	0	0.0%
Pragmatic Play	15	7	46.7%
PragmaticPlay (alias)	1	1	100.0%
Provider blank	1	0	Not calculated
TOTAL	100	40	40.0% of all records

INTERPRETATION

Evolution has 20 missing RTP values across 20 received records (100%). This is a strong reason to investigate the Evolution feed specification or source extract. It does not establish fault or prove that every game in this category should carry an RTP value.

Data handling observations

- The source provider field includes both Pragmatic Play and PragmaticPlay, so canonicalisation is needed before provider-level reporting.
- EV_0036 appears under the received label PragmaticPlay despite an EV identifier and Evolution title. Provider-level counts above intentionally use the received value; a verified correction could change the distribution.
- The blank-provider record PP_0089 is excluded from provider-specific denominators. It is still included in the total feed count.

7. Cleaning and revalidation

The working dataset retained the original fields and added separate proposed/cleaned columns. The original feed was not overwritten.

FIELD	SOURCE VALUE	WORKING VALUE / ACTION	CONTROL
game_id (Play'n GO duplicate)	HG_0010	PG_0011 proposed	Pending identifier verification
provider (PP_0089)	Blank	Pragmatic Play provisional	Pending verification
provider (EV_0036)	PragmaticPlay	Evolution proposed	Pending verification
game_type (HG_0067)	Slots	Slot	Reference-based standardisation
rtp_percent (HG_0028)	0.9624	96.24 proposed	Pending source verification
release_date (BG_0078)	31/31/2024	CHECK / unresolved	Do not invent date
market_codes (BG_0054)	EE;GB;XX	Flagged invalid	Clarify XX

Revalidation outcomes

VALIDATION FIELD	RESULT AFTER WORKING-COPY CHECK
Proposed game IDs	100 unique proposed IDs
Provider names	100 values matched the canonical provider list after proposed cleaning
Game types	99 valid; 1 missing
Volatility	99 accepted values; 1 blank (N/A retained as an accepted label for the exercise)
RTP	60 numeric values passed range/format check; 40 pending applicability review
Market codes	99 valid; 1 pending invalid code
Release dates	99 valid parsed dates; 1 invalid date pending correction
Mobile support	100 accepted Yes/No values
Status	100 accepted values
Provider/game-type compatibility	99 passed; 1 pending due to missing game type

8. Handover assessment and deliverables

RECORD STATUS	COUNT	INTERPRETATION
PASS (automated validation)	56	Passed the combined automated checks in the working copy.
HOLD - REVIEW	44	At least one automated check remained unresolved.
READY FOR HANDOVER	52	Passed the chosen automated criteria and had no external verification flag.
HOLD - DATA QUALITY	44	Held due to data-quality findings.
HOLD - VERIFICATION	4	Passed automated checks but still carried an external verification flag.

IMPORTANT

The handover labels are simulation outcomes generated by workbook rules. They are not production release approvals. A value can pass a formatting or lookup rule and still require provider confirmation.

Prepared workbook outputs

- Incoming Feed: preserved source records and original metadata.
- Provider Reference and Market Reference: lookup tables used during validation.
- Data Dictionary: field definitions and basic validation notes.
- Issue Log: issue IDs, affected fields, evidence, proposed actions and status.
- Cleaned Dataset: working copy with proposed values and validation helper columns.
- Validation Summary: consolidated check results.
- Exception Queue: records held for further investigation.
- Game Master Dataset: 52 records meeting the simulated handover rules.

9. Operational recommendations

Enforce identifier uniqueness: Check incoming game IDs against the current database and the provider's authoritative identifier mapping. Do not silently invent replacement IDs.

Maintain canonical provider reference data: Store canonical provider names, accepted aliases, provider codes and supported categories in a controlled reference table.

Define field applicability: Document when RTP, volatility and other fields are required, optional or not applicable for each game type/provider feed.

Validate every value in multi-value fields: Split market and feature lists into individual elements before validating them against approved references.

Use type-aware validation: Distinguish blank strings, null values, N/A, numeric values, dates and Boolean representations.

Preserve raw source and audit trail: Keep original feed values, proposed corrections, evidence, verification status and update history.

Use an exception queue: Separate records that can proceed from records requiring clarification, provider response or internal approval.

Reconcile every recurring feed: Classify each incoming record as new, unchanged, changed or held for review before updating the central database.

10. Project limitations and integrity

This is an independent training project using synthetic records. Real provider names are included only to make examples recognisable. The dataset does not reproduce a real provider feed, actual commercial integration, verified game catalogue, real RTP certification or legally approved market availability.

All proposed corrections remain provisional unless explicitly supported by authoritative source data. In particular, the suggested replacement identifier PG_0011, provider mappings for PP_0089 and EV_0036, the conversion of HG_0028 RTP to 96.24, and the resolution of the invalid date and market code require external confirmation.

Interview summary

60-SECOND VERSION

I built a simulated iGaming game-metadata workflow in Excel using a 100-record multi-provider feed. I inspected the schema, used reference tables and Pivot Tables to investigate data-quality problems, created traceable proposed corrections, automated validation checks and separated records ready for simulated handover from exceptions. The project taught me to distinguish between detecting a discrepancy and verifying the correct value before updating a central database.